coordinates of the two points), and since this is smaller than the distance to D, our current NN, then we need to visit the other branch of D as well. When we say visit here, we mean traversing the tree in a path from D to the closest leaf to P. While we do so, we visit node F and we discover that it's closer than D, so we update our current NN (and its distance: you can see that we shrink the radius of our nearest neighbor perimeter, the circle marking the region where possible nearest neighbors can be found).

Are we done now? Not yet, we need to keep back-tracking to the root. We go back to node C, but its split line is further away than our NN perimeter (and it didn't have a right branch anyway), so we go back to node A, as shown in figure 9.22.

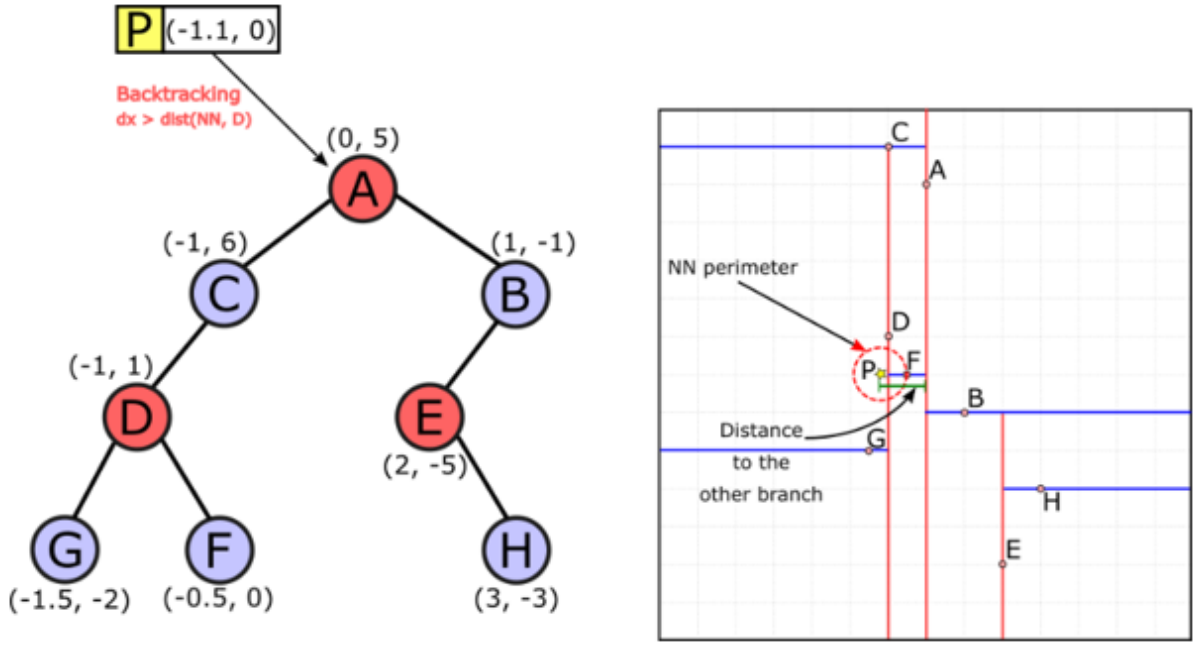


Figure 9.22 We need to backtrack back towards the root. If we had to check every possible branch of the tree, however, this would be no better than scanning